

Atharva Nikam

Mechatronics Engineer (M.Sc.)

Email: atharvaajaynikam4601@gmail.com

Links: [LinkedIn](#) | [Github](#)

Phone: +49 17671744876

Location: 88250 Weingarten, Baden-Württemberg, Germany



PROFESSIONAL SUMMARY

Mechatronics Master's student with hands-on experience in EV powertrain and battery system simulation, prototype development, laboratory testing, and engineering documentation gained through industrial and university projects. This combination of MATLAB/Simulink-based modeling, functional validation with oscilloscopes and multimeters, and a CAD/FEM-based mechanical design background from prior industry roles is what makes me a strong fit for hands-on engineering roles that span simulation, hardware, and documentation. *[Tailor per JD: contribution to this specific role + availability per the JD added per application – master stays generic, no JD to anchor these to.]*

EDUCATION

Sep 2024 – Present	M.Sc. in Mechatronics <i>Ravensburg-Weingarten University of Applied Sciences, Germany</i> Relevant courses: Advanced Hybrids, Electric Drives, Power Electronics
Dec 2020 – Jul 2023	B.E. in Mechanical Engineering <i>Bharati Vidyapeeth College of Engineering, India</i> Relevant courses: Machine Design, Product Design & Development, Manufacturing Processes
Aug 2017 – Nov 2020	Diploma in Mechanical Engineering <i>Bharati Vidyapeeth College of Engineering, India</i> Relevant courses: Elements of Machine Design, Automobile Engineering

PROFESSIONAL EXPERIENCE

Oct 2024 – Present	Student Mentor <i>H.O.M.E. Makerspace (RWU), Germany</i> - Guided student engineering projects from CAD design to prototype fabrication, assembly, electrical integration, and functional testing using laboratory equipment including oscilloscopes and multimeters. - Reduced repeated faults and improved testing efficiency across 20+ engineering projects by introducing structured documentation and systematic troubleshooting methods. - Maintained laboratory equipment and measurement instruments to ensure safe operation and reliable experimental results across workshop activities.
Feb 2024 – June 2024	Computer-Aided Design Engineer <i>Chakra Technologies, Navi Mumbai, India</i> - Developed mechanical components and assemblies using SolidWorks while creating engineering drawings, BOMs, and technical documentation for prototype manufacturing. - Reduced design iterations by evaluating component performance through FEM structural analysis in ANSYS Workbench and collaborating with engineering teams on optimization. - Delivered complete manufacturing documentation packages supporting production planning and engineering communication.
July 2023 – Jan 2024	Graduate Trainee Engineer <i>LCT Pvt. Ltd., Navi Mumbai, India</i> - Improved testing equipment reliability under 2–10 ton operating loads by introducing 5 design modifications based on engineering evaluation. - Reduced workshop preparation time by approximately 2 weeks by collaborating with manufacturing teams during assembly and commissioning. - Developed CAD assemblies and manufacturing drawings for industrial testing equipment, translating design requirements into production-ready documentation.
Sep 2022 – Nov 2022	Junior Faculty Intern <i>DCIT, India</i> - Delivered practical CAD and engineering training covering technical drawings, GD&T, and engineering documentation using industry-standard workflows. - Reviewed 40+ CAD models and engineering drawings, identifying inconsistencies and improving technical documentation quality. - Contributed to a 92% first-attempt certification success rate by improving students' documentation accuracy and engineering practices.

PROJECTS

Predictive Electric Vehicle Cooling – Repo | *Mathworks Capstone Project (#194) – MATLAB & Simulink Challenge Project – Repo* | MATLAB, Simulink, Simscape Modelling

- Integrated longitudinal vehicle dynamics (rolling resistance, aerodynamic drag, motor torque limits, regenerative braking) with a SOC-based battery electrical model in MATLAB/Simulink to simulate EV energy and thermal behaviour under realistic driving and charging conditions.
- Computed dynamic battery thermal loads from irreversible ohmic and reversible entropic heat generation across traction, regenerative braking, and DC fast-charging phases, using a custom 2000s multi-phase mission profile with 25–38°C ambient variation.
- Built a Simscape Fluids thermal management plant – cold plate, thermal-liquid pump, radiator, reservoir, temperature sensing – interfaced with Simulink control signals to model closed-loop battery cooling.
- Developed and compared reactive vs. preview-based predictive cooling controllers, evaluating temperature-constraint violations, cooling-system energy use, and allowable fast-charging power envelope.

MATLAB-Based EV Powertrain Simulation and Data Analysis – Repo | *Independent Personal Project* | MATLAB

- Built a MATLAB-based simulation model of an EV powertrain, modeling battery electrical and thermal behaviour and energy management strategies to analyze system-level performance under varying operating conditions.
- Implemented algorithm-based SOC estimation (Kalman filter), thermal modelling, and regenerative braking logic, processing simulated sensor data to evaluate power electronics and battery system performance.
- Validated simulation outputs against standard drive-cycle data, achieving SOC estimation accuracy within 1.29%.

Multi-Purpose Agricultural Vehicle | *Bharati Vidyapeeth College of Engineering* | AutoCAD, SolidWorks

- Developed the structural chassis architecture and mechanical transmission layouts for a multi-functional utility vehicle prototype.
- Reduced overall vehicle weight by approximately 12% by collaborating with the fabrication team on structural design improvements during manufacturing and assembly.
- Validated prototype performance through field testing, confirming functional operation within 5% of original design calculations.

BAJA Collegiate Motorsports | *Bharati Vidyapeeth College of Engineering* | AutoCAD, SolidWorks

- Designed drivetrain mounting components and mechanical assemblies using SolidWorks while considering manufacturability and assembly requirements.
- Collaborated with team members to integrate mechanical subsystems and improve overall vehicle assembly efficiency.
- Prepared assembly documentation that enabled successful prototype integration several days ahead of schedule.

CORE COMPETENCIES & TECHNICAL SKILLS

Simulation & Programming: MATLAB, Simulink, Simscape Fluids, System Simulation, Data Processing, ANSYS Workbench (Static Structural).

Automotive & Power Systems: EV Powertrain Modeling, Battery SOC Estimation, Power Electronics Fundamentals (coursework: Electric Drives, Power Electronics, Advanced Hybrids).

Engineering Tools: SolidWorks, CATIA V5, AutoCAD, Engineering Documentation, BOM Management.

LANGUAGES & INTERESTS

Languages: English (Fluent - B2/C1), German (Intermediate - B1), Marathi (Native Speaker - C2), Hindi (Fluent - C1)

Interests: Football (*FIFA World Cup & Champions League*), Drawing B&W Sketches (*Human Portraits, Movies & Anime Characters*), Photography (*Human Portraits, Landscapes, Wildlife*), PC Gaming (*Story based*), Evening walk

Weingarten, August 20, 2026



Atharva Nikam